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While an ASIC may consume more power per unit die size than an FPGA, the power is normally amortised over a higher density solution, which provides better power efficiency—Manisha Mankar, Architect-Digital Design, Robert Bosch Engineering and Business Solutions

With this advantage, short design cycle and reconfigurable options, FPGAs are pushing ASICs out of the market."

The fact that ASICs are built for specific applications allows them to have a very high density of useful logic gates on the chip and use resources optimally. Hence higher gate count and lower power consumption give ASICs a competitive edge over FPGAs.

"All the internal structures/IPs are used for customer-specific or mission-specific applications or functions. So, while an ASIC may consume more power per unit die size than an FPGA, power is normally amortised over a higher-density solution, providing better power efficiency," adds Mankar.

Finally, high-volume production

Generally, ASICs are deemed to be the best option for high-volume production.

"If you are really looking for high volumes, FPGAs are out of the game. The cost and performance advantages that you get out of ASICs definitely tilts the balance towards ASICs. FPGAs come into the play when volumes are small and time-to-market is critical. Also, the clock speed is limited in FPGAs. So for high-volume or cost-sensitive applications, ASIC is

probably the way to go," says Milind Gandhi, associate vice president, Semiconductor Business Line, SASKEN.

There is, however, a catch. "The choice of ASIC vs FPGA for volume production is on case-to-case basis. In mobile applications, it makes more sense to have an ASIC-based solution due to the requirement of high density/performance. Time-to-market is also a parameter in such consumer applications. With newer shorter ASIC development cycles, and with less focus on quality, ASICs are the logical choice. On the other hand in network applications, where typically the requirement is moderate in terms of complexity and density, FPGA solutions are preferred due to shorter time-to-market. Quality could be addressed through flexibility in soft-hardware bug fixes," explains Regupathi.

"Generally, for lower production volumes, FPGAs may be more cost-effective than ASICs. But the recent trend suggests that FPGAs could be a better alternative in high-volume applications since costs related to ASIC development are increasing on account of more complex technology nodes. However, at this juncture, it is not very clear whether this trend is sustaining or a temporary aberration," adds Mankar.

For high-volume production, costs associated with building a custom ASIC

Myth Buster: Can FPGAs scale to speeds required by today's Internet?

"There are skeptics in the industry who feel that FPGAs cannot match the speed requirements of today's applications such as Internet routers. However, with changing economics and performance enhancement, FPGA technology is seeking a place to address complex applications. As examples, we do see companies shipping line-rate, high-performance routers using FPGAs," says Vijayakumar Regupathi, architect-digital design, Robert Bosch Engineering and Business Solutions. FPGAs in this context act as a programmable network processor.